

templates\learn\what-is-cwe.html

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{% extends "base.html" %}

{% block head %}
<style>
/* Custom scrollbar styling for dark theme */
/* Creates consistent dark-themed scrollbars across webkit browsers */
.cwe-list-scroll {
max-height: 550px; /* Prevents excessive vertical space usage */
overflow-y: auto; /* Enables vertical scrolling when content exceeds height */
list-style: none; /* Removes default bullet points */
padding: 0; /* Removes default list padding */
margin: 0; /* Removes default list margins */
}

/* Webkit browsers (Chrome, Safari, Edge) scrollbar customization */
.cwe-list-scroll::-webkit-scrollbar {
width: 8px; /* Thin scrollbar for modern appearance */
}

.cwe-list-scroll::-webkit-scrollbar-track {
background: #1a2236; /* Dark background matching site theme */
border-radius: 4px; /* Rounded edges for polished look */
}

.cwe-list-scroll::-webkit-scrollbar-thumb {
background: #63a4ff; /* Blue accent color matching site branding */
border-radius: 4px; /* Consistent rounded appearance */
border: 1px solid #263159; /* Subtle border for definition */
}

.cwe-list-scroll::-webkit-scrollbar-thumb:hover {
background: #5292e6; /* Slightly darker blue on hover */
}

.cwe-list-scroll::-webkit-scrollbar-thumb:active {
background: #4a80cc; /* Even darker when actively dragging */
}

/* Firefox scrollbar styling fallback */
.cwe-list-scroll {
scrollbar-width: thin; /* Thin scrollbar option */
scrollbar-color: #63a4ff #1a2236; /* Thumb and track colors */
}

/* CWE list item styling for interactive elements */
.cwe-list-item {
padding: 8px 24px; /* Comfortable click target size */
cursor: pointer; /* Indicates clickable element */
font-weight: 500; /* Medium weight for readability */
color: #e3e7f2; /* Light gray text for contrast */
user-select: none; /* Prevents accidental text selection */
border-left: 4px solid transparent; /* Space for selection indicator */
transition: background 0.13s, border-left 0.13s, color 0.13s; /* Smooth hover effects */
}

.cwe-list-item:hover {
background: #212d46; /* Darker background on hover */
color: #ffffff; /* Pure white text for maximum contrast */
}

.cwe-list-item.selected {
background: #212d46; /* Same background as hover state */
border-left: 4px solid #63a4ff; /* Blue left border for selection indicator */
color: #63a4ff; /* Blue text to match selection theme */
}

.cwe-code {
color: #94a3b8; /* Muted color for CWE codes */
font-size: 0.9em; /* Slightly smaller than main text */
}

/* Enhanced button styling with hover effects */
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.toggle-button {
margin: 16px 0 0 24px; /* Consistent spacing with list items */
background: #263159; /* Dark blue background */
color: #63a4ff; /* Blue text color */
padding: 8px 18px; /* Comfortable button size */
border-radius: 8px; /* Rounded corners */
border: none; /* Remove default button styling */
cursor: pointer; /* Indicates interactive element */
font-weight: 600; /* Bold text for emphasis */
font-size: 0.97em; /* Slightly smaller than body text */
transition: background 0.2s, transform 0.1s; /* Smooth hover animations */
}

.toggle-button:hover {
background: #2d3d6b; /* Lighter blue on hover */
transform: translateY(-1px); /* Subtle lift effect */
}

.toggle-button:active {
transform: translateY(0); /* Returns to normal position when clicked */
}
</style>
{% endblock %}

{% block content %}
<!-- Main container with responsive max-width -->
<div style="max-width:1400px;margin:0 auto 32px auto;padding-top:20px;">
<!-- Page title with consistent blue theming -->
<h1 style="color:#F5E8D8;">What is a CWE?</h1>
<!-- Comprehensive introduction explaining CWE concept and importance -->
<p style="color:#e2e8f0;font-size:1.13em;">
<strong>CWE</strong> stands for <em><span style="color:#d3bb7a">Common Weakness
Enumeration</span></em>. <br> In
technology and computer science, we learn by identifying patterns?not just memorizing a
list of past mistakes. The <b>CWE catalog</b> is a map of the <span
style="color:#d3bb7a;">most common
error patterns</span> that lead to vulnerabilities in software.<br>
<br>
<b>If you spot the pattern, you can prevent the problem:</b>
<!-- Educational bullet points explaining CWE value proposition -->
<ul style="color:#e2e8f0;margin-left:20px;">
<li>Just like a math book teaches "types of mistakes" (e.g., sign error, division by
zero), CWE teaches you <span style="color:#d3bb7a;">recurring security
problems</span>.</li>
<li><b>CWE entries</b> are like chapters in a textbook?each describes a different kind
of software weakness.</li>
<li>You'll see real-world examples (<b>CVEs</b>), but you'll learn the general idea with
CWE.</li>
</ul>
</p>

<!-- Educational section explaining CWE structure -->
<h2 style="color:#cdd7ff;">CWE Structure: How Weakness Types Are Organized</h2>
<ul style="color:#e2e8f0;margin-left:20px;">
<li><b>ID format:</b> <code>CWE<Number> ? <Name></code> (e.g., <code>CWE-79
? Cross-Site Scripting</code>)</li>
<li><b>Name:</b> A concise label for the type of mistake (e.g., "Buffer Overflow")</li>
<li><b>Description:</b> Explains what usually goes wrong?and why</li>
<li><b>Example:</b> Shows a typical code or configuration that triggers this
weakness</li>
<li><b>Mitigations:</b> Teaches you practical ways to avoid making this mistake when
writing software</li>
<li><b>Hierarchy:</b> Shows if a weakness is broad ("Injection") or a sub-type ("SQL
Injection")</li>
</ul>

<!-- Interactive chart section for data visualization -->
<h2 style="color:#cdd7ff;margin-top:26px;">CWEs by Severity</h2>
<div style="background:#1a2236;border-radius:12px;padding:28px 18px 25px 18px;
margin-bottom:18px;">
<!-- Context information about data source -->
<div style="color:#94a3b8; font-size:0.98em; margin-bottom:9px;">
<span>Showing data from <b>recent analysis</b>. Click any CWE below for live MITRE
data.</span>
</div>

<!-- Filter controls for chart customization -->

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<div style="display:flex;align-items:center; justify-content:space-between;
margin-bottom:18px;">
<label style="color:#e2e8f0;font-size:1.07em; font-weight:600;">
Show:
<!-- Dropdown to control number of displayed CWEs -->
<select id="cweSeverityFilter" style="background:#232b45;color:#63a4ff;padding:7px
14px;border-radius:6px;border:none;margin-left:7px;">
<option value="10" selected>Top 10 CWEs</option>
<option value="5">Top 5 CWEs</option>
</select>
</label>
</div>

<!-- Chart container with responsive wrapper -->
<div style="max-width:1400px;width:100%;overflow-x:auto;">
<canvas id="cwesBySeverityChart" width="700" height="235"></canvas>
</div>
</div>

<!-- JavaScript for chart functionality -->
<script>
// Wait for DOM to be fully loaded before initializing
document.addEventListener('DOMContentLoaded', function() {
// Initialize CWE severity data with fallback values
window.cweSeverityData = window.cweSeverityData || {
labels: ['Cross-Site Scripting', 'SQL Injection', 'Input Validation', 'Path Traversal',
'Buffer Overflow'],
indices: ['CWE-79', 'CWE-89', 'CWE-20', 'CWE-22', 'CWE-119'],
data: {
CRITICAL: [15, 12, 8, 5, 7],
HIGH: [25, 20, 15, 10, 12],
MEDIUM: [30, 25, 20, 15, 18],
LOW: [10, 8, 12, 8, 6]
}
};

// Navigation function to filter vulnerabilities by CWE
function goToVulnsForCWE(id) {
const cwe_key = window.cweSeverityData.indices[id];
if (cwe_key) {
// Redirect to vulnerabilities page with CWE filter
window.location.href = "/vulnerabilities?q=" + encodeURIComponent(cwe_key);
}
}

// Main chart rendering function
function renderChart(topN) {
const data = window.cweSeverityData.data || {};

// Slice data arrays to show only requested number of items
const labels = window.cweSeverityData.labels ? window.cweSeverityData.labels.slice(0,
topN) : [];

// Debug logging for development
console.log("Rendering chart with topN:", topN);
console.log("Labels:", labels);
console.log("Data:", data);

// Chart.js configuration object
const chartData = {
labels: labels,
datasets: [
// Each severity level as separate dataset for stacking
{label:'Critical', data: data.CRITICAL ? data.CRITICAL.slice(0, topN) : [],
backgroundColor:'#f55855'},
{label:'High', data: data.HIGH ? data.HIGH.slice(0, topN) : [],
backgroundColor:'#f8a541'},
{label:'Medium', data: data.MEDIUM ? data.MEDIUM.slice(0, topN) : [],
backgroundColor:'#3b8ded'},
{label:'Low', data: data.LOW ? data.LOW.slice(0, topN) : [], backgroundColor:'#42d392'}
]
};

// Destroy existing chart to prevent memory leaks
if (window.cweSeverityChart) {
window.cweSeverityChart.destroy();
}
}

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// Create new Chart.js instance
window.cweSeverityChart = new
Chart(document.getElementById('cweBySeverityChart').getContext('2d'), {
type: 'bar', // Stacked bar chart type
data: chartData,
options:{
responsive:true, // Adapts to container size
maintainAspectRatio:false, // Allows height customization
plugins:{
legend:{display:true, position:"bottom"}, // Legend below chart
title:{display:true, text:`CWES by Severity ? Top ${topN}`} // Dynamic title
},
scales:{
// X and Y axis configuration with stacking enabled
x:{stacked:true, ticks:{color:'#cdd7ff', font:{size:11}}},
y:{stacked:true, beginAtZero:true, ticks:{color:'#cdd7ff'}}
},
// Click handler for navigation to vulnerability details
onClick: (evt, activeEls) => {
if (activeEls && activeEls.length > 0) {
goToVulnsForCWE(activeEls[0].index);
}
}
});

// Initialize with default Top 10 view
renderChart(10);

// Event listener for filter dropdown changes
document.getElementById('cweSeverityFilter').addEventListener('change', function(){
renderChart(parseInt(this.value,10)); // Parse selected value and re-render
});
});
</script>

<!-- Pass server-side data to JavaScript global variables -->
<script>
// CWE severity data for chart rendering
window.cweSeverityData = [{ cwe_severity | tojson }];
// Complete CWE dictionary for reference
window.learnCweDict = [{ cwe_dict|tojson }];
// Curated list of important CWES
window.key_cwes = [{ key_cwes|tojson }];
// Display names for key CWES
window.key_cwe_titles = [{ key_cwe_titles|tojson }];

// Debug logging for development and troubleshooting
console.log("CWE Severity Data:", window.cweSeverityData);
</script>

<!-- External JavaScript module for CWE interaction handling -->
<script src="{ { url_for('static', filename='js/pages/learn.js') } }"></script>
</div>

<!-- Navigation back to main dashboard -->
<div style="text-align:center; margin:2.5em 0 1em 0;">
<a href="{ { url_for('main.index') } }" style="color:#F5E8D8; padding:0.8em 2em;
border-radius:6px; font-weight:600; text-decoration:none;">
? Back to Dashboard
</a>
</div>
{% endblock %}

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